

Fault Tree Analysis of IUD Perforation

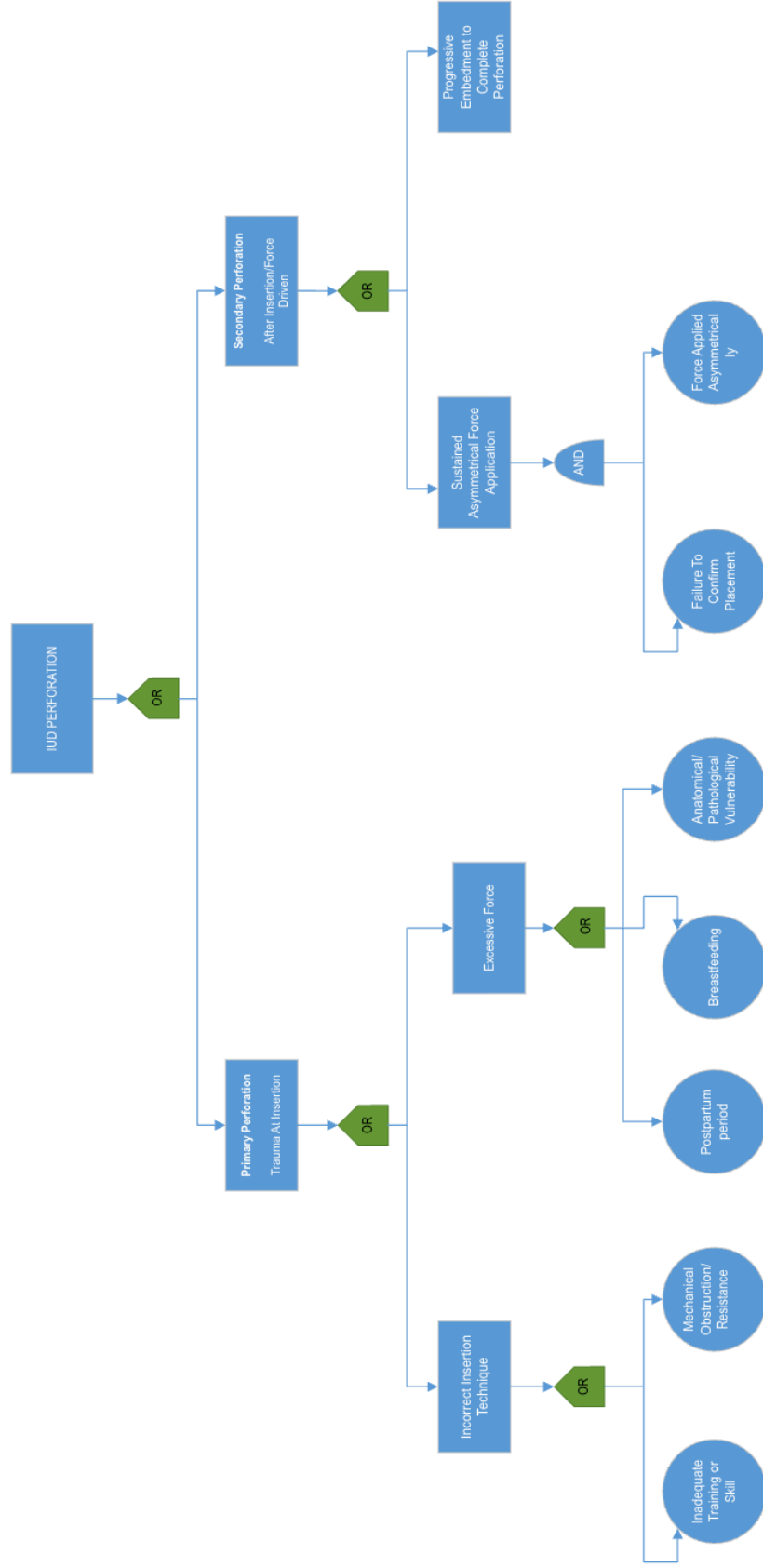
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MSHF 641 Human Factors in Complex Systems

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October 12, 2025



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Intrauterine devices (IUDs) are among the most effective and widely used long acting reversible contraceptives (LARCs). IUDs are designed as small, T-shaped medical devices inserted into the uterus, functioning primarily by altering the uterine environment to prevent fertilization and implantation (Friedman & Oluronbi, 2019). IUDs are a highly effective and safe form of reversible contraception, with an efficacy rate of over 99% in preventing pregnancy (Lanzola et al., 2025). Although serious side effects are rare, one of the most serious complications is uterine perforation, a medical condition where the device penetrates partially or completely through the uterine wall. IUD perforation typically occurs during insertion, but can also develop from device migration or uterine wall weakening after insertion, with the risk of uterine perforation occurring at an estimate of 1 to 2 per 1000 insertions (Rowlands et al., 2016; Anthony et al., 2021).

Women who experience IUD perforation may be asymptomatic, with cases going undetected for months or years. However, symptoms can range from mild to severe, such as abdominal pain, cramping, vaginal bleeding, pain during intercourse, and fever (Rowlands et al., 2016).

In one case of IUD perforation, a 33-year-old woman was admitted to the hospital after a routine examination found a displaced IUD that was implanted six years prior; the woman experiencing no symptoms or discomfort (Li et al., 2024). In another case, after an abdominal X-ray was performed on a 29-year-old woman experiencing moderate lower abdominal pain, it was found that her IUD had moved out of the uterus and into the abdomen (Torres-Cepeda & Reyna-Villasmil, 2016).

Method

Although IUD perforation cases are rare, understanding the mechanisms and contributing factors of IUD perforation is essential for clinicians, device manufacturers, and regulatory agencies. Through systematic methods such as Fault Tree Analysis (FTA), healthcare professionals can understand the interactions between the human, device, and system level failures that can lead to perforation. This analytical approach can enhance patient safety, inform training, device design, and procedural guidelines to minimize future risk. FTA in healthcare is a required element of risk management under ISO 14971:2019, an international standard that establishes a systematic process for medical device manufacturers to identify, evaluate, and control risks associated with their products throughout their lifecycle (Schirn, 2023). Furthermore, FTA in medical devices is often qualitative, emphasizing hazard identification and mitigation over strict numeric probability targets, with each event assessed for potential human harm. To complete an FTA, the analysis must first define the top level event, specifying the undesirable failure to be investigated. The top level event is the starting point for the entire process, as the fault tree is constructed by tracing backwards from the failure to find its root causes.

Analysis

Intermediate Events

IUD perforation can occur at primary perforation or secondary perforation. Primary perforation is caused immediately at insertion, with the device being forced into or through the uterine wall, caused by a failure of insertion technique or excessive force (Goldstuck & Wildemeersch, 2014). Secondary perforation occurs after the initial insertion process, as a result of sustained asymmetrical force application or progressive embedment to complete perforation

(Rowlands et al., 2016). In sustained asymmetrical force application, the uterus generates sufficient force to drive the device out of position, gradually eroding the device through the uterine wall. In progressive embedment to complete perforation, the IUD migrates through the uterine wall over a period of time.

Basic Events

When primary perforation is caused by incorrect insertion technique, this can be attributed to a clinician with inadequate skill or training, leading to incorrect insertion angles or an inability to account for difficult cases. Another cause of incorrect insertion technique is mechanical obstruction and resistance, leading to a complex procedure and a difficult passage through the cervix. While excessive force can be a consequence of inadequate training, it is classified as an intermediate event and connected as an OR gate in this FTA to acknowledge its capacity to be an independent event. Excessive force in this context can occur when the force required for insertion exceeds the weakened resistance of the uterus. This incompatibility in uterus resistance and force can be attributed to patients recently in postpartum state, patients that are breastfeeding at time of insertion, and patients with anatomical vulnerabilities. Patients are found to have a higher risk of IUD perforation during the period of 4 days to 6 weeks postpartum when the myometrium is softer and vulnerable to being torn or punctured (Rowlands et al., 2016). Breastfeeding is also associated with diminished myometrial collagen, thinning, and reduced pliability. Furthermore, patients with undiagnosed structure disorders of the uterus can lead to vulnerability (Tabatabaei et al., 2024).

When secondary perforation is caused by sustained asymmetrical force application, the powerful and repeated force generated by uterine contractions, on a device that is asymmetrical or improperly anchored, can lead to the IUD being driven out of the uterine cavity (Goldstuck &

Wildemeersch, 2014). Secondary perforation can also be caused by progressive embedment to complete perforation, the process by which an IUD slowly erodes, working its way through the uterine wall then migrating from its intended position inside the uterus to a location in the abdominal or pelvic cavity (Rowlands et al., 2016).

Conclusion

Fault Tree Analysis is a powerful and structured method for identifying events that can lead to system failures, particularly within complex sociotechnical environments. Its logical, top down structure helps clarify how component failures, human errors, and environmental conditions interact to produce adverse events. FTA can support risk prioritization and the development of targeted interventions to improve safety and reliability. However, FTA has its limitations, including the nature of its linear framework that limits its ability to capture nonlinear, adaptive, and emergent behaviors characterizing modern healthcare systems. While FTA can reveal critical vulnerabilities, it cannot identify or mitigate every potential hazard in complex sociotechnical systems. These systems evolve dynamically through continuous human, organizational, and technological interactions that produce unpredictable outcomes. Therefore, the goal of safety analysis is not total risk elimination, but resilient risk management such as acknowledging uncertainty, designing for recovery, and fostering a culture of continuous improvement. Integrating FTA with broader systems thinking can help organizations better anticipate, adapt to, and mitigate progressive failures in high impact systems such as medical device safety.

References

- Anthony, M. S., Reed, S. D., Armstrong, M. A., Getahun, D., Gatz, J. L., Saltus, C. W., Zhou, X., Schoendorf, J., Postlethwaite, D. A., Raine-Bennett, T., Fassett, M. J., Peipert, J. F., Ritchey, M. E., Ichikawa, L. E., Lynen, R., Alabaster, A. L., Merchant, M., Chiu, V. Y., Shi, J. M., Xie, F., ... Asimwe, A. (2021). Design of the Association of Uterine Perforation and Expulsion of Intrauterine Device study: a multisite retrospective cohort study. *American Journal of Obstetrics and Gynecology*, 224(6), 599.e1–599.e18.
<https://doi.org/10.1016/j.ajog.2021.01.003>
- Friedman, J., & Oluronbi, R. A. (2019). Optimizing IUD Delivery for Adolescents and Young Adults. Springer, Cham. https://doi.org/10.1007/978-3-030-17816-1_3
- Goldstuck, N. D., & Wildemeersch, D. (2014). Role of uterine forces in intrauterine device embedment, perforation, and expulsion. *International Journal of Women's Health*, 6, 735–744. <https://doi.org/10.2147/IJWH.S63167>
- Lanzola, E. L., Auber, M., & Ketvertis, K. (2025). Intrauterine Device Placement and Removal. StatPearls Publishing.
- Li, Q., Qi, D., Bi, T., Guo, X., & Chen, H. (2024). Case report: Uterine perforation caused by migration of intrauterine devices. *Frontiers in Medicine*, 11, 1455207.
<https://doi.org/10.3389/fmed.2024.1455207>
- Rowlands, S., Oloto, E., & Horwell, D. H. (2016). Intrauterine devices and risk of uterine perforation: current perspectives. *Open Access Journal of Contraception*, 7, 19–32.
<https://doi.org/10.2147/OAJC.S85546>

Schirn, A. (2023). ISO 14971:2019–Risk Management To Medical Devices. American National Standards Institute.

<https://blog.ansi.org/ansi/iso-14971-2019-risk-management-to-medical-devices>

Tabatabaei, F., Hosseini, S. T. N., Hakimi, P., Vejdani, R., & Khademi, B. (2024). Risk factors of uterine perforation when using contraceptive intrauterine devices. *BMC Women's Health*, 24(1), Article 538. <https://doi.org/10.1186/s12905-024-03298-3>

Torres-Cepeda, D., & Reyna-Villasmil, E. (2016). Small bowel perforation by an intrauterine device. *Gastroenterología y Hepatología (English Edition)*, 39(7), 495–496. <https://doi.org/10.1016/j.gastre.2016.06.011>